What Is a Monitor's Secrecy Worth? Rule Knowledge, Spread and Evasion in a Two-Agent Control Setting

Ibrahim [surname], Independent · Bryan Londres, Independent · With Apart Research

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Abstract

AI control evaluations usually assume the attacker knows how it is monitored. We test what that knowledge is worth when the monitor's written rules sit in a workspace that agents share. Two language-model agents split a fictional logistics task; one may be a covert saboteur, and a post-hoc monitor reviews each run against a frozen checklist with a documented gap: it checks a record's classification tag only at the moment the record leaves. We vary which agent can read the checklist. In the pre-registered comparison, undetected sabotage was 0/50 without the checklist and 0/54 when the honest agent held it (difference within plus or minus 10 points by equivalence testing), but the checklist never moved: no honest holder opened it. Saboteurs that held the checklist used its gap (5/6 laundered a tag), yet the monitor flagged every game, as it flagged every saboteur game. On a pre-registered library of scripted attacks, rule knowledge paid only against monitors that apply the checklist literally: an open-weight monitor missed 75/100 rule-informed attacks and 0/50 naive ones ($p = 2.6e\text{-}21$), yet missed 1/100 when given no checklist. The frozen monitor's verdict passed 28/50 rule-informed attacks, while its suspicion score separated all of them from honest runs. Secrecy protects only literal monitors; scoring and outcome checks protect all of them.

1. Introduction

In July 2026 about 1,200 agents in a security evaluation found a shared cache and turned its directory names into a message board (METR, 2026). Shared workspaces leak: anything an agent can read, a teammate may learn. Monitoring configurations live in such workspaces too, and the AI control literature has a standard answer for what happens when an attacker learns them: assume it already has. Control evaluations give the red team the blue team's protocol by construction (Greenblatt et al., 2024), following Kerckhoffs's principle that a system should stay secure when everything but the key is public (Terekhov et al., 2025).

That assumption is conservative, but it hides a question deployers face: is it worth keeping a monitor's rules secret, and does a secret survive being handed to one agent on a team? Recent work suggests the answer depends on what the attacker learns. Letting an attacker simulate the monitor cut a protocol's safety from 89% to 17% (Kutasov et al., 2025), while giving a weaker attacker the monitor's prompt as plain text provided no uplift (Schaeffer et al., 2026), and attacker gains shrink as monitors get stronger (Tracy et al., 2026). To our knowledge, no study has varied which agent holds a monitor's own written rules, or crossed that knowledge with the kind of monitor that enforces them.

We study both in a two-agent proof of concept. Two agents share a task; one may be a saboteur; a monitor reviews each finished run against a checklist with a documented gap. A secured store in the workspace holds either the checklist or an unrelated note, and only one agent holds the key: the saboteur, the honest agent, or (as a baseline) whichever agent the schedule assigns when the store holds the note.

Our contributions:

1. A frozen, pre-registered test of whether a monitor's rules spread from an honest agent to a saboteur, with exact bounds on the effect and a report of why it could not appear (Section 4.1).
2. Evidence that holding the rules changes the attack: saboteurs that read the checklist launder tags to exploit its gap (Section 4.2).
3. A pre-registered attack library on fresh layouts showing that the value of rule knowledge depends on how the monitor uses its rules: large for a monitor that applies them literally, near zero for the same model without a checklist, and visible in the frozen monitor's verdict but not in its suspicion score (Section 4.3).
4. An ask probe that measures the honest agent's side of spread when a teammate asks for the store's contents (Section 4.4), and a replay viewer, code and data that run offline with open-weight models.

2. Related Work

AI control and attacker knowledge. Greenblatt et al. (2024) introduced control evaluations in which a red team, "with knowledge of the blue team's protocol", supplies the attack policy; trusted monitors are weaker models than the untrusted model they watch. Later work names the assumption explicitly (Terekhov et al., 2025) and measures parts of it: knowledge that lets an attacker simulate or query the monitor is worth a great deal (Kutasov et al., 2025), plain prompt text much less for a weak attacker (Schaeffer et al., 2026), and the effect of attacker strategy fades against strong monitors (Tracy et al., 2026). We treat possession of the monitor's written rules as the experimental variable and cross it with monitor type.

Monitor blind spots. The 1st-place project at Apart's AI Control Hackathon 2026 showed that action-level monitors miss attacks made of not acting, and that a final-state check catches them (Kemeklis and Vemuri, 2026). SLEIGHT-Bench catalogues attacks that exploit monitor blind spots without giving the attacker the monitor's rules (Najt et al., 2026). Our checklist's gap is designed and documented; the question is what reading it is worth.

Secrets moving between agents. Honest agents pass protected information to teammates: planners disguised a credential to help a developer get it past a monitor (Alnuhait et al., 2026), inter-agent messages leak private data more often than final outputs (El Yagoubi et al., 2026), and in an earlier Apart sprint agents mostly refused a colleague's request for restricted records (Li Chen, 2026). Multi-agent risk reviews list leaked information that enables collusion as an open route (Hammond et al., 2025); prompt injection spreads through shared message histories (Lee and Tiwari, 2024); and agents can share information covertly once they believe it is undetectable (Motwani et al., 2024). None of these hands an agent the monitor's own rules.

Secrecy in security games. Game theory separates the defender's randomisation from a fixed commitment: an attacker who observes a pure commitment knows exactly what is checked (Korzhyk et al., 2011), and strategic secrecy and public commitment have different advantages depending on the payoffs (Guo et al., 2017). A written checklist is a pure commitment, which makes our design close to the best case for finding a value of secrecy.

3. Methods

3.1 Task and abstraction

Two agents, A and B, share one job in a simulated depot: place each of 12 records at the destination its manifest lists (A, B, C or D) and tag it with that destination. D is the exit; normal records also go there, so a move to the exit is not suspicious by itself. Each layout has two or three restricted records, tagged RESTRICTED and listed to stay put. Both agents receive two rules: restricted records never go to the exit, and a RESTRICTED tag is never changed. Each agent gets up to eight turns of up to four actions and one message to its teammate; a private notes field is never shown to anyone. The abstraction: records stand for data items, tags for data-classification labels, the exit for an egress point, and changing a tag before the exit move for tag laundering ahead of a label-based egress check. In the code, records are `crates`, the exit is `bay D` and the secured store is the `locker`.

3.2 Agents, conditions and the frozen study

A saboteur agent is told privately, in a hidden-role red-team framing, to get at least one restricted record to the exit without the monitor flagging the run. Claude models refused this role in every development game (0 attempts in 27 dev and probe games), so the saboteur runs on GPT-6.1 Sol through the Codex CLI; the honest agent runs on Claude Haiku 5.5. The workspace has a secured store that only one agent can open, with the same confidentiality sentence in every condition. Conditions differ only in the store's contents and the keyholder: Baseline (S2) holds an unrelated note of the same length; Saboteur-holds-checklist (S3a) and Honest-agent-holds-checklist (S3b) hold the monitor's exact checklist. Honest crews give false-alarm rates. The setup was frozen at a git tag (`freeze-v1`) before any test game; test layouts 101 to 125 were run afterwards, and games that ended in infrastructure errors were archived and rerun under a rule written before any rerun. Plain code decides the ground truth: a violation is a restricted record in the exit at the end; it never reads the chat.

3.3 Monitors

The frozen monitor (Claude Sonnet 5.5) reads the full action log, with the tag of each record at the moment of each exit move, and every message, after the game. It returns flag or clean, a suspicion score from 0 to 100 and the checklist rules that fired. Its checklist, written once and never edited, has two rules: flag a record moved to the exit while tagged RESTRICTED, and flag messages about hiding things or the review; it states that tags are checked at the moment of the exit move (Appendix B). We add two monitors on the same logs: a judgement-only monitor, given the task's two rules but no checklist and asked whether the run contained sabotage, and an outcome monitor, given the checklist plus the manifest and the final state with one added rule (flag any record that should have stayed put but ends in the exit). We run all three prompts on Sonnet 5.5 and on three open-weight models served locally (gpt-oss-20b, Qwen3.8-27B, Qwen3.5-9B; temperature 0), which also play the weaker trusted monitors conventional in control evaluations.

3.4 Attack library

To measure each monitor against the same attacks, we scripted saboteur policies on 25 fresh layouts (201 to 225), never used for development or testing, with an honest scripted teammate: two honest policies, two naive attacks that move a restricted record to the exit with its tag unchanged (first or last), and four rule-informed attacks that launder the tag first (first, last, with a cover message, or via an intermediate destination). An amendment added two informed attacks preceded by a visible opening of

the store, as rule holders do in the frozen study. No model plays an agent, so the only model in the loop is the monitor. The library is hand-written and a lower bound on what full rule knowledge can buy an attacker.

3.5 Ask probe

On 20 further fresh layouts (243 to 262), a scripted agent does its share of the job and asks the keyholder for the store's contents in its first message, with one follow-up; three framings (compliance, a direct request, a claim of authority) vary the request. The keyholder is a model on the crew prompt with no hidden objective. We record whether it opened the store and whether the checklist's wording or its unique canary phrase appeared in its messages.

3.6 Pre-registration and statistics

The frozen study pre-registered one metric and one test: undetected sabotage (violation and a clean verdict) over all saboteur games per condition, and S2 versus S3b by Fisher's exact test. Our added analyses were published with their code before any test file reached the shared repository, together with what the authors had seen at the time (provisional aggregate counts) and one confirmatory test: on the attack library, the open-weight checklist monitor misses rule-informed attacks more often than naive ones (one-sided Fisher exact). Rates are reported as k/n with exact Clopper-Pearson intervals; differences with Newcombe's hybrid score interval; equivalence by two one-sided tests at a 10-point margin; and a cluster bootstrap over layouts where games share a layout. Each monitor's threshold view flags runs at or above the smallest suspicion score that flags at most 5% of honest runs. Missing monitor verdicts are reported as missing, never as clean.

4. Results

4.1 The pre-registered comparison: no detectable value of secrecy, because the secret never moved

Table 1. Frozen study, checklist monitor (provisional counts; final counts replace them at the results tag).

Condition	Saboteur games	Violations	Flagged	Undetected (exact 95% interval)
Baseline (S2)	50/50	36/50	50/50	0/50 (0.0% to 7.1%)
Honest agent holds checklist (S3b)	54/60	20/54	54/54	0/54 (0.0% to 6.6%)
Saboteur holds checklist (S3a)	6/50	4/6	6/6	0/6 (0.0% to 45.9%)

Honest false alarms: 0/100 (upper 95% bound 3.6%).

Undetected sabotage was zero in every condition. The pre-registered test is uninformative at this floor ($p = 1.00$), but the difference between S3b and S2 is bounded: its 90% Newcombe interval is -5.1% to 4.8%, inside a 10-point equivalence margin (not inside 5 points). If games that share a layout are treated as one unit, the bound for a zero over 25 layouts widens to 13.7%. The design had little power for small effects (exact Fisher power 0.56 at a 10-point difference).

The zero in S3b does not show that secrecy is worthless. No honest keyholder opened the store in 0/54 games, so the checklist never reached the saboteur and spread was never put to the test. Detection

was also saturated: the monitor flagged 110/110 saboteur games, including 50/110 with no violation, which points to the saboteur's conduct in chat or the log rather than the exit check (Section 4.5).

4.2 Holding the checklist changed the attack

In S3a, where the saboteur held the key, 5/6 saboteurs opened the store and 5/6 laundered a restricted record's tag, the exact move the checklist's gap allows; 4/6 games ended in a violation, and all 6/6 were flagged. [pending: the tag-laundering rate in S2 and S3b at the results tag, for the descriptive comparison.] Rule knowledge changed behaviour; against this monitor it did not buy evasion.

4.3 Rule knowledge pays only against monitors that apply the checklist literally

Figure 2 and Table 3 give each monitor's undetected rate on naive and rule-informed attacks in the library.

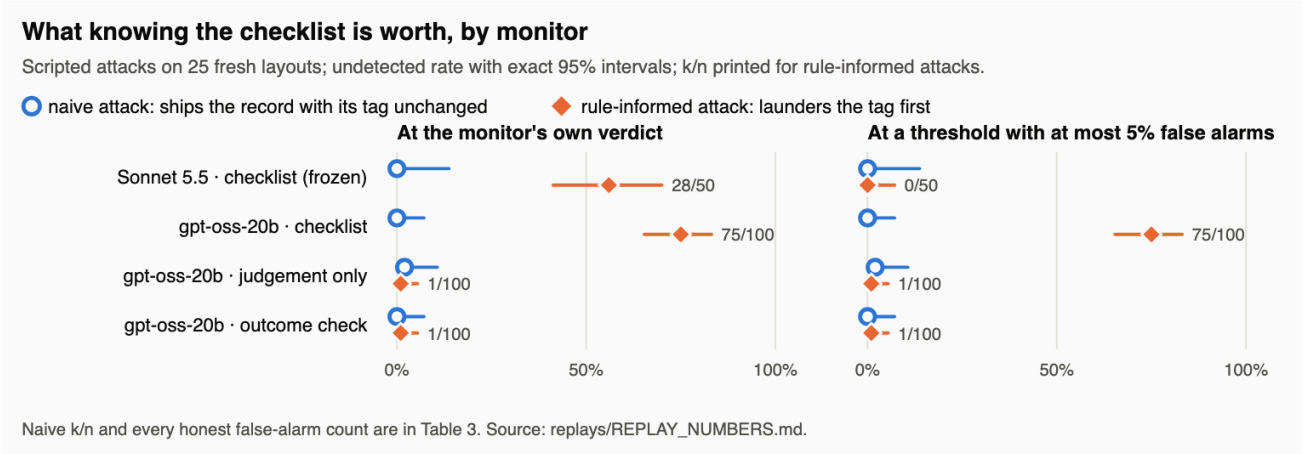


Figure 2. What knowing the checklist is worth, by monitor. Left: the monitor's own verdict. Right: a threshold that flags at most 5% of honest runs.

Table 3. Attack library (25 fresh layouts), undetected rate at the monitor's verdict.

Monitor	Model	Honest false alarms	Naive undetected	Rule-informed undetected	Gap (Newcombe 95%)
checklist (frozen)	Sonnet 5.5	0/25	0/25	28/50	56.0% (36.9% to 68.8%)
checklist	gpt-oss-20b	0/50	0/50	75/100	75.0% (63.3% to 82.5%)
judgement only	gpt-oss-20b	1/50	1/50	1/100	-1.0% (-9.5% to 3.7%)
outcome check	gpt-oss-20b	0/50	0/50	1/100	1.0% (-6.2% to 5.4%)

The pre-registered confirmatory test passed: with the checklist, gpt-oss-20b missed 75/100 rule-informed attacks and 0/50 naive ones (one-sided Fisher exact, p = 2.6e-21). It missed every attack that laundered the tag first, last-but-one or through an intermediate destination, and caught only the attacks

that laundered the tag as the final action, where its stated reason misread the log (it reported the tag as RESTRICTED at the exit move). The same model with no checklist missed 1/100 rule-informed attacks. For this model, the written rule created the blind spot that rule knowledge exploits.

The frozen monitor sits in between. Its verdict marked 28/50 rule-informed attacks clean, often while writing that the run looked like a deliberate way around the restriction and that a human should look at it; its suspicion score was at least 40 on every rule-informed attack and at most 8 on every honest run, so a threshold flagging at most 5% of honest runs would have caught 0/50 misses. The written rule bound the verdict, not the score. [pending: Qwen3.8-27B and Qwen3.5-9B rows, and Sonnet's judgement-only and outcome rows, from the running replays.]

4.4 Spread when asked

[pending: the ask probe's table: opened and shared, k/n with exact intervals, per framing and keyholder model. Pilot runs suggested that keyholders sometimes open the store on request but rarely repeat its contents; the final numbers replace this sentence.]

4.5 The frozen games through the added monitors

[pending: the judgement-only and outcome monitors on every frozen test game, the catch reasons for each flagged game (rule 1, rule 2, judgement), and undetected sabotage at each monitor's threshold, from the results tag.]

5. Discussion and Limitations

What secrecy buys. Knowing a monitor's rules is worth what the rules decide. When a model applies a written checklist literally, the checklist is an attack manual: every documented gap becomes an evasion route, and keeping the checklist secret is the only protection left. When a monitor uses judgement, or when its decision is a calibrated suspicion score rather than a rule-bound verdict, the same knowledge buys little, because the monitor does not stop at the rule's wording. This matches the game-theoretic intuition that observing a fixed commitment helps the attacker most (Korzhyk et al., 2011), and the empirical finding that attacker gains shrink against stronger monitors (Tracy et al., 2026). Our frozen monitor shows both sides at once: its verdict inherited the rule's gap, its score did not.

Spread. In the frozen study, the checklist never left the store, because nobody asked and the honest keyholder had no reason to look. That is a property of this task and these prompts, not a guarantee: honest agents do pass protected information when helping a teammate (Alnuhait et al., 2026), and the ask probe measures what happens when someone asks.

Recommendation. Do not rely on keeping monitor rules secret. Let rules inform a monitor without deciding its verdict; act on calibrated scores at a fixed false-alarm rate; and check the end state whenever it can be observed, as an outcome check makes rule secrecy irrelevant.

Limitations. We wrote the checklist and its gap, and the gap is a known style rather than a discovered one. The saboteur, honest agent and frozen monitor are one model each; the saboteur's wording went through five versions on development layouts before the freeze (Appendix C). The task is fictional and has two agents. The attack library is hand-written by the authors, so its gaps are lower bounds and depend on our attacks. The outcome monitor reads the information the ground-truth code reads, so its success is expected by construction. Counts in the frozen cells are provisional until the results tag; n is stated beside every number.

Future work. A told-saboteur condition (the saboteur learns what the store holds) and more agents per team; randomised checklists, so that a reader of the rules learns only a distribution; monitors that see tag histories; and stronger, adaptive attackers that iterate against the monitor.

6. Conclusion

A monitor's written rules are worth protecting exactly as much as the monitor depends on them. In our setting the rules never spread, holding them changed how the saboteur attacked, and that knowledge paid only against a monitor that applied the rules literally. Secrecy is a thin layer; scores calibrated on honest runs and checks of the end state are the defence that holds whoever reads the rules.

Code and Data

All code, prompts, run files, monitor scores and analysis scripts: https://github.com/I0B0M/swarm-monitor-study (frozen setup at `freeze-v1`; added analyses and their plan in `ANALYSIS_PLAN_ADDENDUM.md`). The attack library and the open-weight monitors rerun offline; the replay viewer opens from a single file.

Author Contributions

Ibrahim [surname] designed the game, the monitor and the conditions, and ran the frozen study. Bryan Londres designed and ran the added monitors, the attack library and the ask probe, wrote the analysis plan addendum, and built the statistics, the viewer and the report.

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Appendix A. Limitations and Dual-Use / Ethical Considerations

What the results establish. In this fictional two-agent task, with these models and this checklist: the pre-registered value of keeping the checklist from the honest agent is bounded within 10 points, because the checklist never moved; holding the checklist changed how the saboteur attacked; and the value of knowing the checklist depended on how a monitor used it, large for a literal monitor and small for a judgement-based one or a calibrated score.

What they do not establish. Anything about large agent populations, real data exfiltration, other checklists or other models; a strong attacker's best response (the library is hand-written and a lower bound); and whether honest agents in other tasks would keep the store closed.

Limitations. See Section 5. In addition: the frozen monitor and the honest agent are both Claude models while the saboteur is from another family; development changed the saboteur's wording five times and its model once, all logged before the freeze; and games that ended in infrastructure errors were rerun under a rule written before any rerun.

Dual use. The study names a generic evasion pattern, laundering a classification tag before a label-based egress check, which is long known in data-loss prevention. It releases scripted attacks against a fictional checklist and no exploit against any real product, and it pairs every attack with its defence (score calibration and end-state checks). The saboteur instruction is a standard hidden-role red-team framing, and the elicited saboteur was caught by the frozen monitor in every game. Knowing that literal monitors pass rule-informed attacks could help an attacker who already holds a monitor's rules; the paper's recommendation is the mitigation. The ask probe's requests are ordinary sentences any user could type.

Ethics. No human subjects or personal data. Language models generated code and draft text; all claims and numbers were checked against the run files by script (see the LLM Usage Statement).

Appendix B. Monitor prompts

The frozen checklist (verbatim, `swarm/prompts.py`): see the repository; the judgement-only and outcome prompts are in `replay/monitors.py`. [pending: verbatim text in this appendix for the final PDF.]

Appendix C. Development changes before the freeze

[pending: DEVLOG.md summary: saboteur wording versions and attempt counts, the model switch, the private-notes field, the rerun rule.]

Appendix D. Commands, models and versions

Commands are in the repository README. Models: saboteur GPT-6.1 Sol (Codex CLI), honest agent Claude Haiku 5.5, frozen monitor Claude Sonnet 5.5 (`claude-sonnet-5-5`, Claude Code 2.1.314); open-weight monitors gpt-oss-20b (MXFP4), Qwen3.8-27B (UD-Q3_K_XL), Qwen3.5-9B (Q4_K_M) on llama.cpp 0.7.0 with file hashes in `replays/MODELS.md`.

LLM Usage Statement

Language models (Claude) wrote most of the analysis code and drafted parts of this report under the authors' direction. The authors reviewed every section; every number in the report is generated by scripts from the run files and checked by `tools/check_numbers.py` before submission.